Reinforcement Learning

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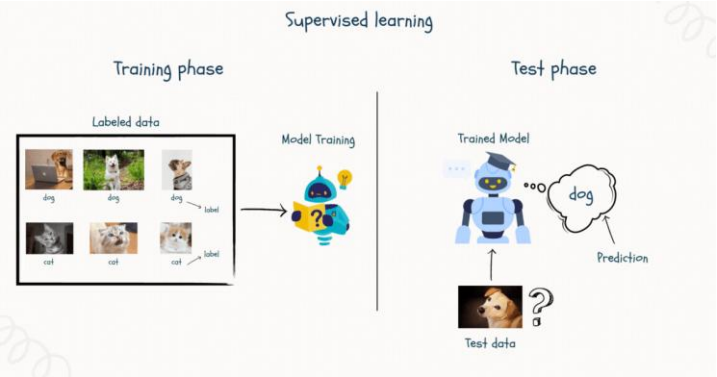
Machine Learning

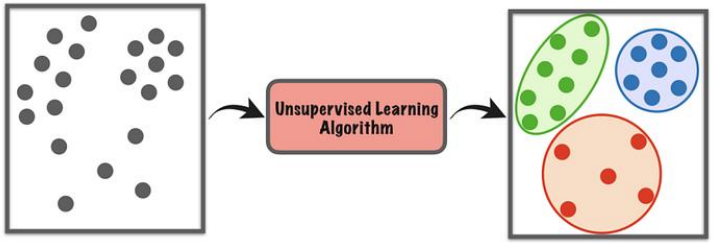
- Supervised learning (지도학습)
  - Learn from labeled data (input + correct answer)
  - Common tasks
    - Regression: predict house prices
    - Classification: spam detection, object detection (car or dog)
  - Algorithms
    - Linear regression, decision trees, neural networks
- Unsupervised learning (비지도학습)
  - Learn from unlabeled data (no answers provided)
  - Common tasks
    - Clustering: group similar data
    - Dimensionality reduction: simplify data
  - Algorithms
    - K-means clustering, PCA (principal component analysis)
- Reinforcement learning (강화학습)
  - Learn by interacting with an environment (trial and error)
  - Common tasks
    - Sequential decision-making: game playing (chess), self-driving cars, robotics
  - Algorithms
    - Q-learning, DQN (deep Q networks), Policy gradient methods

Machine Learning

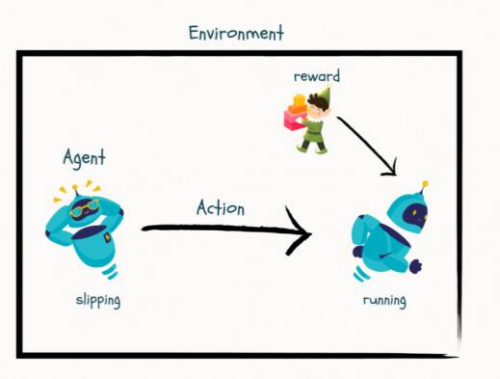
Supervised learning



Unsupervised learning



Reinforcement learning

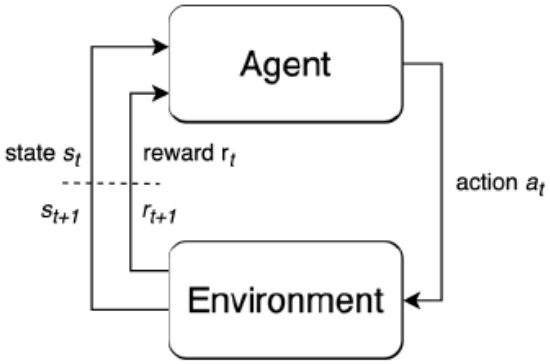


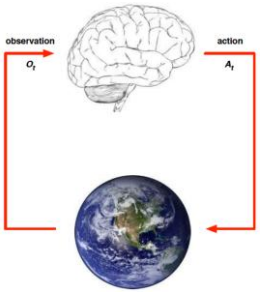
- An **agent** takes actions in an environment.
- Receives **rewards or penalties**.
- Learns a strategy to maximize total reward.

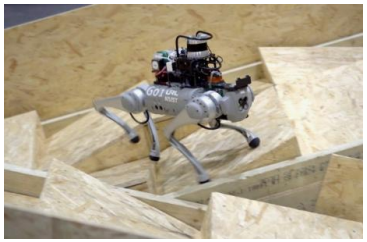
* 강화 (强化) 학습 : <심리학> 보상과 처벌을 통해 최적의 행동을 학습하는 방법

Reinforcement Learning

- Components of RL
  1. **Agent** (학습 주체)
     - Decision-maker (player)
       - It chooses actions based on what it observes
  2. **Environment** (환경)
     - Everything the agent interacts with
       - The environment responds to the agent's action
       - It gives feedback (next state + reward)
  3. **State** (상태)
     - The current situation of the environment
       - Used by the agent to decide what to do next
  4. **Action** (행동)
     - What the agent can do
       - The agent chooses an action based on the current state
  5. **Reward** (보상)
     - Feedback from the environment
       - A number that tells the agent how good it action was
       - The goal is to maximize total reward over time



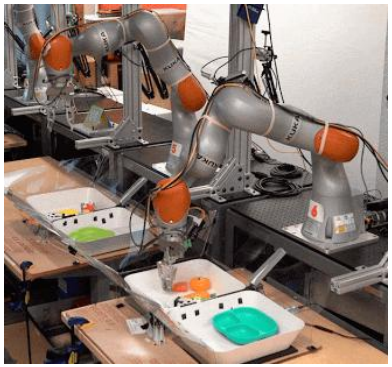


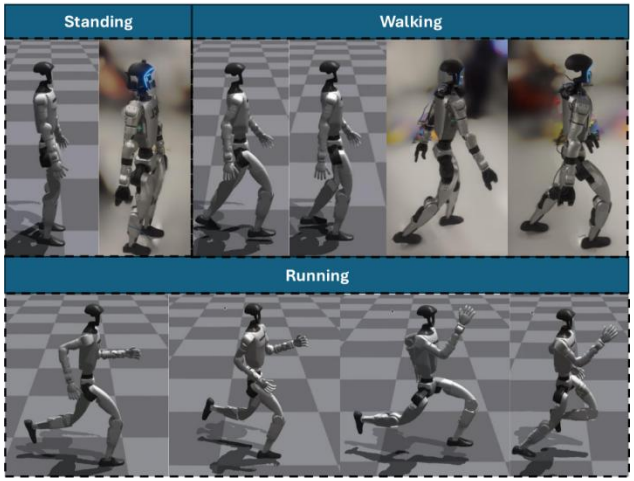


(ex) 로봇 보행 문제
- Agent: 로봇 (제어기)
- Environment: 도로, 장애물
- Action: 각 다리 관절의 이동 각도/속도
- State : 장애물 위치
- Reward: 로봇이 이동한 거리

Common Tasks of RL

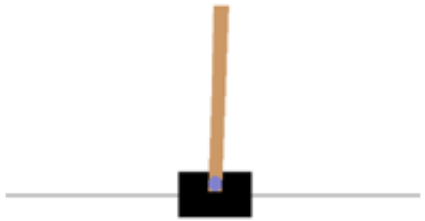
- Sequential decision-making
  - Playing games or sports
  - Self-driving cars
  - Robot manipulation
  - Humanoid locomotion

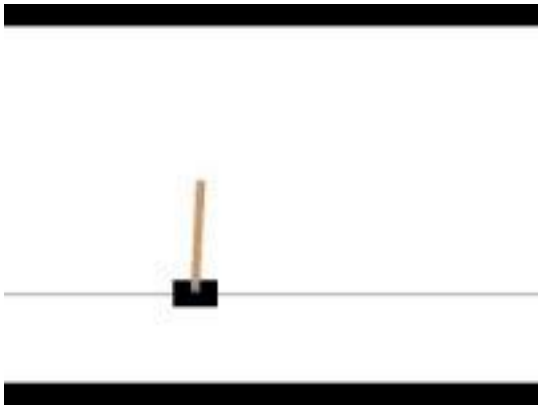




RL Examples

1) CartPole





1. **Objective:** Keep the pole upright for 200 time steps.
2. **State:** An array of length 4 which represents: [cart position, cart velocity, pole angle, pole angular velocity]. For example, [−0.034, 0.032, −0.031, 0.036].
3. **Action:** An integer, either 0 to move the cart a fixed distance to the left, or 1 to move the cart a fixed distance to the right.
4. **Reward:** +1 for every time step the pole remains upright.
5. **Termination:** When the pole falls over (greater than 12 degrees from vertical), or when the cart moves out of the screen, or when the maximum time step of 200 is reached.

https://youtu.be/SLQrHrmAaLk

RL Examples

2) Atari Breakout

1. **Objective:** Maximize the game score.
2. **State:** An RGB digital image with resolution 160×210 pixels—that is, what we see on the game screen.
3. **Action:** An integer from the set $\{0, 1, 2, 3\}$ which maps to the game controller actions {no-action, launch the ball, move right, move left}.
4. **Reward:** The game score difference between consecutive states.
5. **Termination:** When all game lives are lost.

RL Examples

3) Biped Walker (Humanoid Robot)

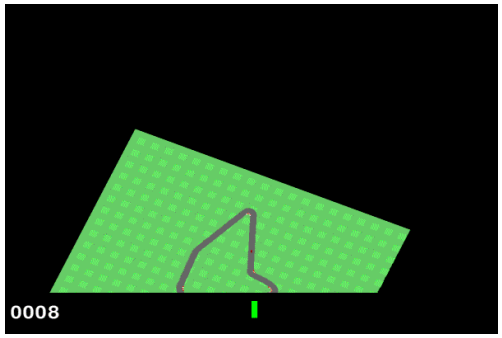
https://youtu.be/p78kRgPZoAA

https://youtu.be/cA37RQKtoU0?t=6

1. **Objective:** Walk to the right without falling.
2. **State:** An array of length 24 which represents: [hull angle, hull angular velocity, x-velocity, y-velocity, hip 1 joint angle, hip 1 joint speed, knee 1 joint angle, knee 1 joint speed, leg 1 ground contact, hip 2 joint angle, hip 2 joint speed, knee 2 joint angle, knee 2 joint speed, leg 2 ground contact, . . ., 10 lidar readings]. For example, [2.745e−03, 1.180e−05, −1.539e−03, −1.600e−02, . . ., 7.091e−01, 8.859e−01, 1.000e+00, 1.000e+00].
3. **Action:** A vector of four floating point numbers in the interval [−1.0, 1.0] which represents: [hip 1 torque and velocity, knee 1 torque and velocity, hip 2 torque and velocity, knee 2 torque and velocity]. For example, [0.097, 0.430, 0.205, 0.089].
4. **Reward:** Reward for moving forward to the right, up to a maximum of +300. −100 if the robot falls. Additionally, there is a small negative reward (movement cost) at every time step, proportional to the absolute torque applied.
5. **Termination:** When the robot body touches the ground or reaches the goal on the right side, or after the maximum time step of 1600.

RL Examples

4) Car Racing



1. Objective
2. State
3. Action
4. Reward
5. Termination

Challenges in RL

- Sparse & delayed feedback
  - SL: you get clear input-output labels
  - RL: you only get rewards, often delayed
- Not fixed & self-generated data
  - SL: dataset is fixed and independent
  - RL: data depends on the agent's action by the trade-off between
    - Exploration: trying new actions
    - Exploitation: using known good actions
- Sequential and interdependent decisions
  - Actions affect future states and rewards
  - The agent must learn a policy over time, not just a single prediction
- Unstable and expensive training
  - Rewards can be noisy or sparse (high variance)
  - Small changes can lead to large behavioral differences (unstable)
  - RL requires interacting with an environment. Real-world data collection can be slow or expensive.

History of RL

- Trial-and-error learning (psychology)
  - Late 1800s ~ Early 1900s
  - reward/punishment shaping behavior
- Dynamic programming
  - 1950s – 1960s
  - R.Bellman, dynamic programming
- Early RL algorithms
  - 1980s – 1990s
  - R. Sutton (1988), Temporal difference learning
  - C. Watkins (1989), Q-learning
- Neural network RL
  - 2000s
- Deep RL
  - 2010s
  - Deep Q-networks (2013) by DeepMind, AlphaGo (2016)
- Modern RL
  - 2020s – present
  - RLHF (RL from human feedback) by OpenAI
  - Physical AI

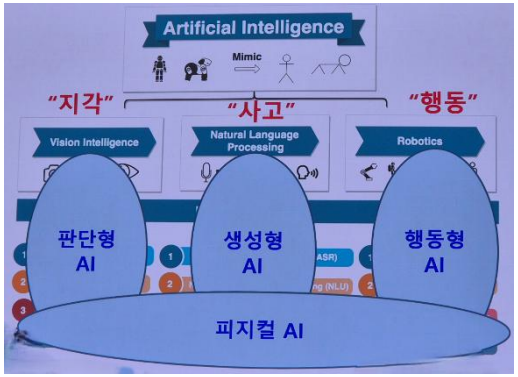
AI 기술의 진화

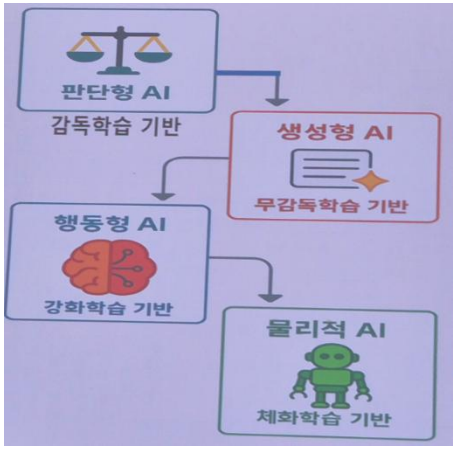
- AI 구분
  - 판단형 AI (지각)
    - Vision intelligence
    - Supervised learning based
  - 생성형 AI (사고)
    - Natural language processing
    - Unsupervised learning based
  - 행동형 AI (행동)
    - Robotics
    - Reinforcement learning based
- AI 의 진화

판단형 AI → 생성형 AI → 행동형 AI

"Learn to Act"

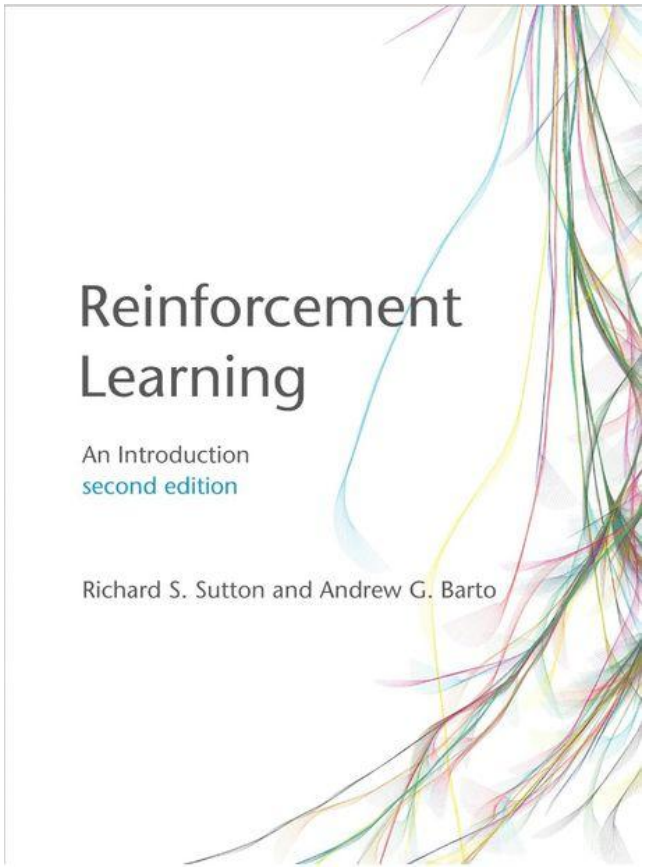
(ICROS 2025)

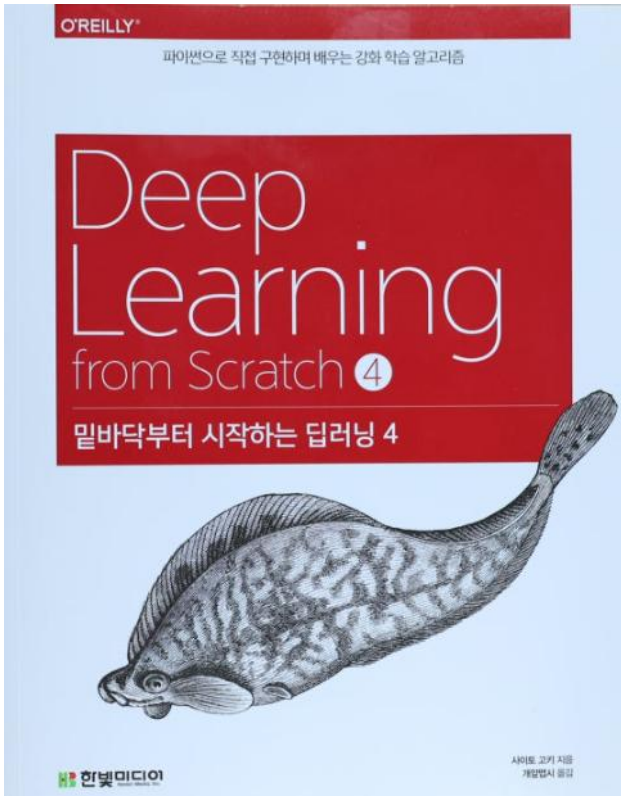




Course Outline

- Textbooks





https://github.com/WegaLee/deep-learning-from-scratch-4

Course Outline

- Contents
  - Introduction : 1 week
  - Markov Decision Process : 1 week
  - Dynamic Programming : 2 weeks
  - Monte Carlo Method : 1 week
  - Temporal Difference Method: 2 weeks
  - Neural Networks: 1 week
  - Q Network: 1 week
  - DQN : 1 week
  - Policy Based Method : 1 week
  - Advanced RL: 1 week
  - Term projects: 1 week

- 성적
  - 시험 (중간 + 기말) : 60%
  - 과제 (Projects + HWs) : 40%
  - 출석: 결석 및 지각 감점, ¼ 결석 시 F 처리